

## A rational exponent

Use real values.

$$x^{3/2} = 27$$

**What solution is permitted by the exponent's domain?**

## An even numerator

Use the real cube root convention.

$$x^{2/3} = 9$$

**Why might this equation have a negative solution as well as a positive one?**

## A fractional-power substitution

Use real values.

$$x^{2/3} - 5x^{1/3} + 6 = 0$$

**What substitution turns this into a quadratic equation?**

## A factored polynomial equation

$$(x + 1)(x - 2)(x - 4) = 0$$

**What are all the real solutions?**

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## Find a common factor

$$x^3 - 9x = 0$$

**What are all the real solutions?**

## Factor by grouping

$$x^3 + 2x^2 - 9x - 18 = 0$$

**What are all the real solutions?**

## A radical equation

A radical equation contains the variable inside a radical.

**What should we check after raising both sides to an even power?**

## Solve a square-root equation

$$\sqrt{2x + 5} = 5$$

**What real solution satisfies the original equation?**

## A radical and a variable

$$\sqrt{x + 6} = x$$

**Which candidates survive substitution into the original equation?**

## A radical has a nonnegative output

$$\sqrt{x+2} = -3$$

**Can any real number satisfy this equation?**

## A shifted radical

$$\sqrt{x + 4} + 2 = x$$

**Which real solutions satisfy this equation?**

## An odd root

$$\sqrt[3]{2x - 1} = -3$$

**What is the real solution?**

## Two radical expressions

$$\sqrt{x+5} - \sqrt{x} = 1$$

**What real solution satisfies the equation?**

## Absolute value as distance

$$|x - 4| = 7$$

**Which points are 7 units from 4?**

## Isolate absolute value

$$3|2x + 1| - 4 = 11$$

**What are all the real solutions?**

## An impossible absolute value

$$|x + 2| = -5$$

**Why is the solution set empty?**

## Quadratic form

An equation in quadratic form is quadratic in some expression.

$$x^4 - 13x^2 + 36 = 0$$

**What substitution reveals the quadratic?**

## Return to the original variable

$$x^4 - 13x^2 + 36 = 0$$

**What are all the real values of x?**

## A quadratic in a shifted expression

$$(x + 1)^2 - 4(x + 1) - 5 = 0$$

**What are all the real solutions?**

## A rational equation with a quadratic

$$\frac{6}{x} + \frac{x}{2} = 5$$

**Which real solutions satisfy the original equation?**

## Keep original restrictions

$$\frac{x^2 - 4}{x - 2} = 5$$

**What solution remains after accounting for the original denominator?**

## Squaring and equivalence

$$x = -2$$

**Does squaring both sides produce an equivalent equation?**

## Graphical solution checks

$$\sqrt{x+6} = x$$

**How would two graphs help us check the number of real solutions?**